Trajecta: A Heterogeneous, Traffic-Aware and Infrastructure-Conscious Smart Logistics Optimization Framework

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Abstract. Routing optimization for logistics networks has grown considerably harder as fleets diversify and cost structures extend beyond simple distance. This paper presents **Trajecta**, an extension of the earlier *Profecta* system that combines heterogeneous fleet modeling, three-dimensional (3D) capacity validation, traffic-aware travel-time estimation, infrastructure feasibility checking, stochastic service-time modeling, and a seven-factor cost model within a single heuristic AI pipeline. Eight order-splitting strategies and seven traffic-aware routing heuristics generate 56 candidate routes per order batch, from which three Pareto-representative routes (minimum cost, minimum distance, minimum penalty) are surfaced for dispatcher review. On a synthetic 30-order, 18-city instance, Trajecta's best-found route achieves a 31% total-cost reduction and up to a 100% penalty reduction relative to a Gurobi-CVRP baseline restricted to distance-and-penalty costing, with 95% on-time delivery versus 62% for the baseline; part of this gap reflects the richer cost model rather than the routing algorithm alone, and all figures are specific to this synthetic instance.

Keywords. Routing Optimization, Heuristic Search, Mixed Fleets, Stochastic Model, Vehicle Routing Problem

1. Introduction

E-commerce growth has pushed last-mile and inter-city transportation costs to the forefront of logistics planning [1]. Fuel, operational cost of the transport mode, salary of drivers, penalty for late delivery, premium service cost, and cost of the infrastructure [1] make up a substantial part of the logistics cost. Existing models which solve the routing problems consider distance or travel time to be minimum assuming all the vehicles to be same and traffic to be constant [2].

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In the current scenario, a fleet can be composed of two wheelers to multi-axle diesel vehicles [3]. Delays are time-varying, weather can add significant time to a trip segment, and bridge or road-weight restrictions can rule out a route for a given vehicle entirely [4,5]. Off-the-shelf optimization models generally ignore all of this.

This paper introduces **Trajecta**, an extension of *Profecta* [6] that integrates heterogeneous fleet management, 3D capacity checking, traffic-aware routing, infrastructure feasibility checking, and a seven-factor cost model.

Trajecta targets three operational outcomes: lower fuel cost via better vehicle-order matching, fewer late-delivery penalties via time-sensitive prioritization, and balanced workload via fleet utilization. A geospatial visualization layer lets planners inspect top-ranked routes before dispatch.

Contributions: (i) 3D simultaneous capacity checking prior to vehicle assignment; (ii) multiplicative, multi-factor travel-time modeling; (iii) infrastructure feasibility checks integrated before route scoring; and (iv) stochastic service-time estimation calibrated against published logistics literature.

2. Related Work

VRP research has moved from pure distance minimization toward more realistic formulations: learning-based routing [2]; cost-oriented models combining freight, warehousing, and carbon-budget objectives [8,9]; and metaheuristics—ACO, PSO, and hybrid ACO–Tabu Search—that outperform exact solvers under time constraints [10,11]. Separately, split-delivery VRPs have established complexity benchmarks for order-splitting [12], and 3D-loading-constrained formulations have addressed volumetric packing [13,14].

Gap: No prior system simultaneously combines heterogeneous fleet management with 3D capacity checking, dynamic traffic modeling, hard infrastructure feasibility checks, stochastic service-time estimation, and a seven-factor cost model—the gap Trajecta addresses.

3. Methodology

3.1. Problem Setup and Data

For experiments, we consider a synthetic case study with 30 orders in 18 Indian cities, with parameters (weight, volume, delivery windows, handling indicator) sampled from Indian e-commerce logistics distributions available in literature (no proprietary data involved). This consists of a fleet size of seven (petrol/electric two wheelers, petrol/diesel/electric four wheelers, and diesel/electric trucks). There is an 18×18 matrix of distances between the cities, along with four traffic conditions (rush morning, rush evening, mid-day, and night).

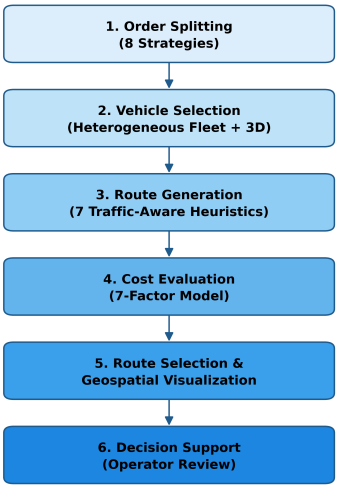


Figure 1. System Architecture and Flow Depiction of Trajecta

3.2. System Architecture

Trajecta converts an unordered order list into ordered routes through six distinct steps (Figure 1): Splitting, allocation of vehicles, creation of routes, calculation of costs, visualization, and dispatcher authorization (Sections 3.3-3.8).

3.3. Order Splitting (8 strategies)

Orders are split by Estimated Delivery Date, Arrival Date, Latitude, Travel Time, Longitude, combined Latitude–Longitude, Penalty-Risk, and Density-Based (DBSCAN) clustering — the last two new, extending the six strategies inherited from Profecta [6] to target financially risky orders and spatially dense clusters respectively.

3.4. Routing Heuristics (7 heuristics)

Within each split, routes are generated by seven heuristics — Nearest Neighbor, Bi-Directional Search, Most Urgent Destination (EDD-ordered), Farthest Insertion, Randomized Nearest Neighbor, Greedy Time-Window, and Cluster-First Route-Second — all beginning/ending at the depot, giving $8 \times 7 = 56$ candidate routes evaluated in Section 5.

3.5. Traffic-Aware Travel Time

$$T_{ij}^{v,t} = \frac{d_{ij}}{s_v} \times \lambda_v \times \omega \times \tau_t \tag{1}$$

where d_{ij} is segment distance, s_v vehicle base speed, λ_v a lane factor, and ω, τ_t weather and time-of-day multipliers. Factors compound multiplicatively rather

than adding, so multiple adverse conditions degrade travel time faster than their sum.

3.6. Stochastic Service Time

$$S_i = t_{\text{base}} + \alpha \cdot w_i + \beta \cdot V_i + \gamma \cdot (f_i + c_i) + \delta \cdot \text{skill}, \text{skill} \sim \mathcal{N}(\mu_{\text{skill}}, \sigma_{\text{skill}}) \quad (2)$$

Service time at each stop depends on order weight w_i, volume V_i, fragility/cooling flags, and a normally distributed worker-skill term. Coefficients ($\alpha = 0.05$ min/kg, $\beta = 5.0$ min/m^3) were set to match aggregate service-time statistics reported in published last-mile logistics literature [15], not fitted to real transaction data — no such data was available for this study.

3.7. Seven-Factor Cost Model

Each route is scored on seven cost components and summed as:

$$\text{TotalCost(route)} = FC + \text{Wages} + PC + cp_cost + m_cost + TC + CC \quad (3)$$

where FC is fuel cost (distance × rate × terrain-gradient factor), $Wages$ is driver pay (distance × rate, plus a supply-chain allowance where applicable), PC is penalty cost (late days × daily rate, summed over orders), cp_cost covers cooling/packaging, m_cost covers maintenance and eco-compliance, TC is toll fees by vehicle class/axle count, and CC is EV charging cost (energy required × tariff). This extends Profecta's five-factor model [6] with toll fees and EV charging cost.

3.8. Algorithm Summary

Combining all splits and heuristics yields the 56 scored candidates described above, from which the three routes (minimum cost, distance, and penalty) are extracted — deliberately avoiding collapse into a single "best" answer so trade-offs remain visible to the dispatcher. The dominant cost is the $\mathcal{O}(N^2)$ heuristic evaluation, repeated $S \times H = 56$ times, giving overall complexity $\mathcal{O}(SH(N^2 + E + N))$, where E is the number of infrastructure checkpoints.

4. Experimental Setup

All 56 candidate routes are evaluated on the 30-order synthetic instance (Section 3.1). Results are compared against a Gurobi-CVRP exact baseline restricted to a simplified distance-and-delay-penalty cost function. The randomized components (Randomized Nearest Neighbor, stochastic service-time sampling) are run 10 times with different seeds; deterministic heuristics produce identical results across runs. Measured wall time is ∼1,250 ms on an Intel i5/8GB machine, versus ∼12,500 ms for Gurobi-CVRP on the same instance.

Table 1. Route optimization results by objective

Objective	Split	Heuristic	Vehicle	Total Cost (Rs.)	Penalty (Rs.)
Min. Cost / Min. Distance	EDD	CFRS	4W-EV	4,200	0
Min. Penalty (alt.)	EDD	MU	2W-EV	4,448	100
Next-best (alt.)	DensityGeo	RNN	2W-Petrol	5,415	100

5. Results and Discussion

Note: minimum-cost and minimum-distance rankings coincide here because the EV route has zero fuel cost and low penalty exposure on this particular instance — this need not hold in general.

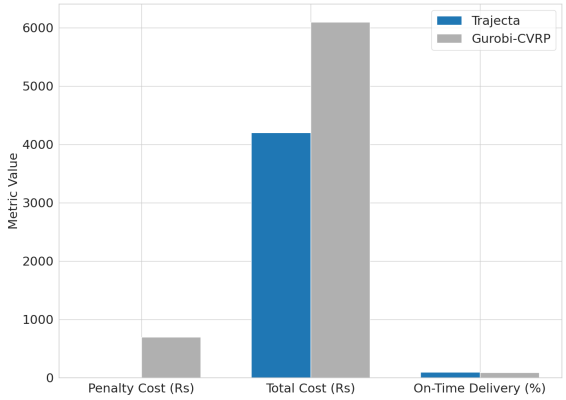


Figure 2. Comparison of Trajecta with Existing Systems

Trajecta's minimum-cost route (Route 7: EDD split, CFRS heuristic, 4W-EV) achieves a total cost of Rs. 4,200 (31% below Gurobi-CVRP's Rs. 6,100) and a penalty cost of Rs. 0 (100% below Rs. 680), by eliminating fuel cost (EV) and avoiding toll segments. Route distances across all 56 candidates clustered between 420–520 km (mean 467 km), with Cluster-First Route-Second averaging 438 km — the best distance/penalty trade-off. On-time delivery for the minimum-cost and minimum-penalty routes reached 95% versus 62% for Gurobi-CVRP. Baseline differences in cost modeling are discussed in Section 6.

6. Limitations

Results rest on a single synthetic 30-order instance, with no real order, fleet, or traffic data, and no optimality gap for the heuristic search. Baseline comparisons mix cost-model and algorithmic effects (Section 5). Service-time coefficients are calibrated against published literature rather than real transaction data, traffic multipliers use four static daily bands, and EV charging assumes a static tariff. Routes are computed offline with no replanning after dispatch. Industry-partner validation is needed before any deployment claim.

7. Conclusion

This paper introduced Trajecta, extending Profecta along four axes: 3D capacity checking, multiplicative traffic-aware travel-time modeling, infrastructure feasibility checking, and stochastic service-time estimation, combined with a seven-factor cost model. On a single synthetic 30-order instance, Trajecta's heuristic pipeline outperformed a distance-and-penalty Gurobi-CVRP baseline on cost, penalty, and on-time delivery, though results are preliminary and instance-specific (Section 6). Future work includes validation with an industry partner, live-traffic integration, learned traffic multipliers, and multi-depot extensions.

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